

## SAFETY DATA SHEET

Creation Date 04-Oct-2010

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Revision Number 3

### 1. Identification

**Product Name** Rhodium(III) chloride hydrate

**Cat No. :** 43702

**CAS No** 20765-98-4  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                   |            |
|-----------------------------------|------------|
| Corrosive to metals               | Category 1 |
| Acute oral toxicity               | Category 4 |
| Serious Eye Damage/Eye Irritation | Category 1 |
| Germ Cell Mutagenicity            | Category 2 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May be corrosive to metals  
Harmful if swallowed  
Causes serious eye damage  
Suspected of causing genetic defects

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Use personal protective equipment as required  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Keep only in original container

**Response**

IF exposed or concerned: Get medical attention/advice

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER or doctor/physician

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Spills**

Absorb spillage to prevent material damage

**Storage**

Store locked up  
Store in corrosive resistant polypropylene container with a resistant liner

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component                   | CAS No     | Weight % |
|-----------------------------|------------|----------|
| Rhodium trichloride hydrate | 20765-98-4 | <=100    |
| Rhodium trichloride         | 10049-07-7 | -        |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

|                                            |                           |
|--------------------------------------------|---------------------------|
| <b>Most important symptoms and effects</b> | Causes severe eye damage. |
| <b>Notes to Physician</b>                  | Treat symptomatically     |

## 5. Fire-fighting measures

|                                         |                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                 |
| <b>Flash Point</b>                      | No information available                                                 |
| <b>Method -</b>                         | No information available                                                 |
| <b>Autoignition Temperature</b>         | No information available                                                 |
| <b>Explosion Limits</b>                 |                                                                          |
| <b>Upper</b>                            | No data available                                                        |
| <b>Lower</b>                            | No data available                                                        |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                 |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                 |

### Specific Hazards Arising from the Chemical

Explosive properties. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
1

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                                                                                                                           |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.                                                                                                                                |

## 7. Handling and storage

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents.                                                           |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component                   | ACGIH TLV                   | OSHA PEL                               | NIOSH                                                     | Mexico OEL (TWA)            |
|-----------------------------|-----------------------------|----------------------------------------|-----------------------------------------------------------|-----------------------------|
| Rhodium trichloride hydrate | TWA: 0.01 mg/m <sup>3</sup> | (Vacated) TWA: 0.001 mg/m <sup>3</sup> | IDLH: 2 mg/m <sup>3</sup><br>TWA: 0.001 mg/m <sup>3</sup> | TWA: 0.01 mg/m <sup>3</sup> |
| Rhodium trichloride         | TWA: 1 mg/m <sup>3</sup>    | (Vacated) TWA: 0.1 mg/m <sup>3</sup>   | IDLH: 100 mg/m <sup>3</sup><br>TWA: 0.1 mg/m <sup>3</sup> |                             |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location.  
Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended Filter type:**

Particulates filter conforming to EN 143.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                         |
|----------------------------------------|-----------------------------------------|
| Physical State                         | Solid                                   |
| Appearance                             | Red brown                               |
| Odor                                   | No information available                |
| Odor Threshold                         | No information available                |
| pH                                     | No information available                |
| Melting Point/Range                    | 100 °C / 212 °F                         |
| Boiling Point/Range                    | No information available                |
| Flash Point                            | No information available                |
| Evaporation Rate                       | Not applicable                          |
| Flammability (solid,gas)               | No information available                |
| Flammability or explosive limits       |                                         |
| Upper                                  | No data available                       |
| Lower                                  | No data available                       |
| Vapor Pressure                         | No information available                |
| Vapor Density                          | Not applicable                          |
| Specific Gravity                       | No information available                |
| Solubility                             | Soluble in water                        |
| Partition coefficient; n-octanol/water | No data available                       |
| Autoignition Temperature               | No information available                |
| Decomposition Temperature              | No information available                |
| Viscosity                              | Not applicable                          |
| Molecular Formula                      | Cl <sub>3</sub> Rh . x H <sub>2</sub> O |
| Molecular Weight                       | 209.26                                  |

## 10. Stability and reactivity

**Reactive Hazard**

None known, based on information available

|                                         |                                                           |
|-----------------------------------------|-----------------------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.                           |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Avoid dust formation. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                                   |
| <b>Hazardous Decomposition Products</b> | Hydrogen chloride gas                                     |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                  |
| <b>Hazardous Reactions</b>              | None under normal processing.                             |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Component Information

| Component                   | LD50 Oral          | LD50 Dermal   | LC50 Inhalation |
|-----------------------------|--------------------|---------------|-----------------|
| Rhodium trichloride hydrate | 753 mg/kg (rat)    | >2 g/kg (rat) | Not listed      |
| Rhodium trichloride         | 1302 mg/kg ( Rat ) | Not listed    | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Risk of serious damage to eyes                                                           |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                   | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------|------------|------------|------------|------------|------------|------------|
| Rhodium trichloride hydrate | 20765-98-4 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Rhodium trichloride         | 10049-07-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** Ames test: positive. Category 2

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                   | Freshwater Algae | Freshwater Fish                     | Microtox   | Water Flea |
|-----------------------------|------------------|-------------------------------------|------------|------------|
| Rhodium trichloride hydrate | Not listed       | Leuciscus idus:<br>LC50=220mg/L/96h | Not listed | Not listed |

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN3260  
 Proper Shipping Name Corrosive solid, acidic, inorganic, n.o.s.  
 Technical Name Rhodium trichloride hydrate  
 Hazard Class 8  
 Packing Group III

#### TDG

UN-No UN3260  
 Proper Shipping Name Corrosive solid, acidic, inorganic, n.o.s.  
 Hazard Class 8  
 Packing Group III

#### IATA

UN-No UN3260  
 Proper Shipping Name Corrosive solid, acidic, inorganic, n.o.s.  
 Hazard Class 8  
 Packing Group III

#### IMDG/IMO

UN-No UN3260  
 Proper Shipping Name Corrosive solid, acidic, inorganic, n.o.s.  
 Hazard Class 8  
 Packing Group III

### 15. Regulatory information

#### United States of America Inventory

| Component                   | CAS No     | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|-----------------------------|------------|------|--------------------------------------------------|--------------------------------|
| Rhodium trichloride hydrate | 20765-98-4 | -    | -                                                | -                              |
| Rhodium trichloride         | 10049-07-7 | X    | ACTIVE                                           | -                              |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

#### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                   | CAS No     | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|-----------------------------|------------|-----|------|-----------|-------|------|------|------|-------|----------|
| Rhodium trichloride hydrate | 20765-98-4 | -   | -    | -         | X     | -    |      | -    | X     | -        |
| Rhodium trichloride         | 10049-07-7 | X   | -    | 233-165-4 | X     | X    | X    | X    | X     | KE-30327 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### U.S. Federal Regulations

#### **SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

#### **SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

#### **CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

### **U.S. State Right-to-Know Regulations**

| Component           | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------------|---------------|------------|--------------|----------|--------------|
| Rhodium trichloride | X             | -          | -            | -        | -            |

### **U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

### Other International Regulations

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH** Not applicable

| Component                   | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Rhodium trichloride hydrate | 20765-98-4 | -                                                                   | -                                                                             | -                                                                                                     |

|                     |            |   |   |   |
|---------------------|------------|---|---|---|
| Rhodium trichloride | 10049-07-7 | - | - | - |
|---------------------|------------|---|---|---|

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                   | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Rhodium trichloride hydrate | 20765-98-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Rhodium trichloride         | 10049-07-7 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component                   | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Rhodium trichloride hydrate | 20765-98-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Rhodium trichloride         | 10049-07-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Creation Date

04-Oct-2010

## Revision Date

29-Mar-2024

## Print Date

29-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**